STAT 391 Final Project

Varun Panuganti, Pavan Vemparala, Ben Epstein, Shuhaib Tait, Gus Elovitz

March 18, 2026

1 Introduction

Our project is about understanding what makes a song popular in the music industry, and we will be using a dataset of Spotify songs to conduct this analysis. This dataset contains 32,000 Spotify tracks, containing features like danceability, energy, loudness, valence, and tempo. We will train regression models (using cross validation and regularization to prevent overfitting) to evaluate which features are the most significant in predicting popularity, analyze correlation between predictors, and observe how our findings differ across control factors like genre. Our main goal is to understand which specific factors affect a songs popularity in a data-driven platform setting like Spotify.

2 Dataset Description

Our dataset contains information for individual songs accessed through the Spotify Application with the `spotifyr` package. The dataset contains 32,833 observations, each of which represents a unique song. The dataset can be found at:

`https://github.com/rfordatascience/tidytuesday/blob/main/data/2020/2020-01-21/readme.md`

We identified the feature `track_popularity` as the most interesting response variable and examined predictors that could explain a song's popularity, or lack thereof.

3 Data Exploration & Preparation

To begin, we first identified a number of largely categorical variables that would not contribute to our analyses. We made the following decisions prior to beginning our investigation:

1. Used `random_state = 314` with `n = 1000` to randomly select a large sample of songs from the dataset. We argue that examining all 32,000+ songs at once renders the experiment unnecessarily complicated. We chose a seed in order to maintain consistency across experiments.

2. Dropped useless categorical variables. We decided that the following variables were unnecessary to our experiment:

    a. `track_id`, `track_name`, `track_album_id`, `track_album_name`, `track_album_release_date`, `playlist_name`, `playlist_id`: these variables contained information independent to each song. While some of these variables might provide micro-level information, perhaps into which albums were more successful, this did not align with the scope of our experiment.

b. `track_artist`: artist would allow us to examine which artists have been the most successful. However, due to the limited number of songs by a single artist across the dataset, and our random sampling, we omitted this field as it would be skewed and not provide any rigorous information.

c. `playlist_subgenre`: we decided that `playlist_subgenre` was too specific and would lead to the creation of too many micro-features that would lead to overfitting.

3. Categorized `playlist_genre`: The genre of song was a feature we felt was highly relevant to our examination. As such, we used the pandas `get_dummies` function to categorize each of our genres, enabling a genre-level examination.

4. Examined collinearity: In our first question we chose to examine the relationships between our remaining features to identify if any collinearity would cause our dataset to be biased.

After this, we were able to begin our preliminary analysis of the dataset. For our initial analysis, we chose to use the whole dataset, even though future experiments would be performed utilizing our random sample.

3.1 Dataset Shape

```
(32833, 23)
```

3.2 Missing Values

```
track_id 0
track_name 5
track_artist 5
track_popularity 0
track_album_id 0
track_album_name 5
track_album_release_date 0
playlist_name 0
playlist_id 0
playlist_genre 0
playlist_subgenre 0
danceability 0
energy 0
key 0
loudness 0
mode 0
speechiness 0
acousticness 0
instrumentalness 0
liveness 0
valence 0
tempo 0
duration_ms 0
```

This dataset is extremely complete and leaves almost no room for error. There is almost no missing data. It is important to note that 5 tracks do not have an album. However, these could

very likely be solo songs. Regardless, album is not a feature of interest to us and thus, this dataset is complete and our observations will not omit any data.

3.3 Popularity Statistics

```
count 32833.000000
mean 42.477081
std 24.984074
min 0.000000
25% 24.000000
50% 45.000000
75% 62.000000
max 100.000000
```

1. Median > Mean: There is a slight left-skew in our dataset, likely caused by songs with a popularity score of 0, or perhaps no true score.
2. SD = 25: A large standard deviation tells us that our songs in the middle are dramatically clustered. Most of the songs are in this range but the range is quite large.
3. Interquartile Range (IQR): The middle 50% of songs score between 24 and 62.
   a. If a song is above 62, it is in the top 25% (a "hit" in this dataset).
   b. If it is below 24, it is in the bottom 25%.

Percentage of songs with 0 popularity: 8.23%

Nearly 1/12 songs in this dataset have a 0 popularity score. While we do not have any true missing data, this is the closest to missing data we have. Likely, songs accumulate popularity as they continue to grow but early songs, or unreleased songs, will remain at a popularity score of 0.

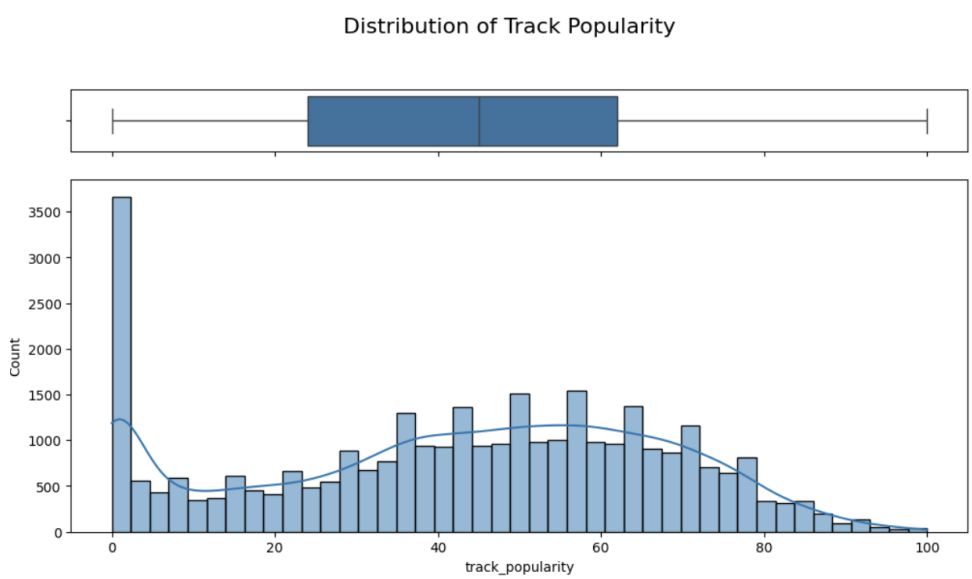

Figure 1: Distribution of track popularity.

4 Question 1: Which predictors in this dataset are heavily correlated to each other?

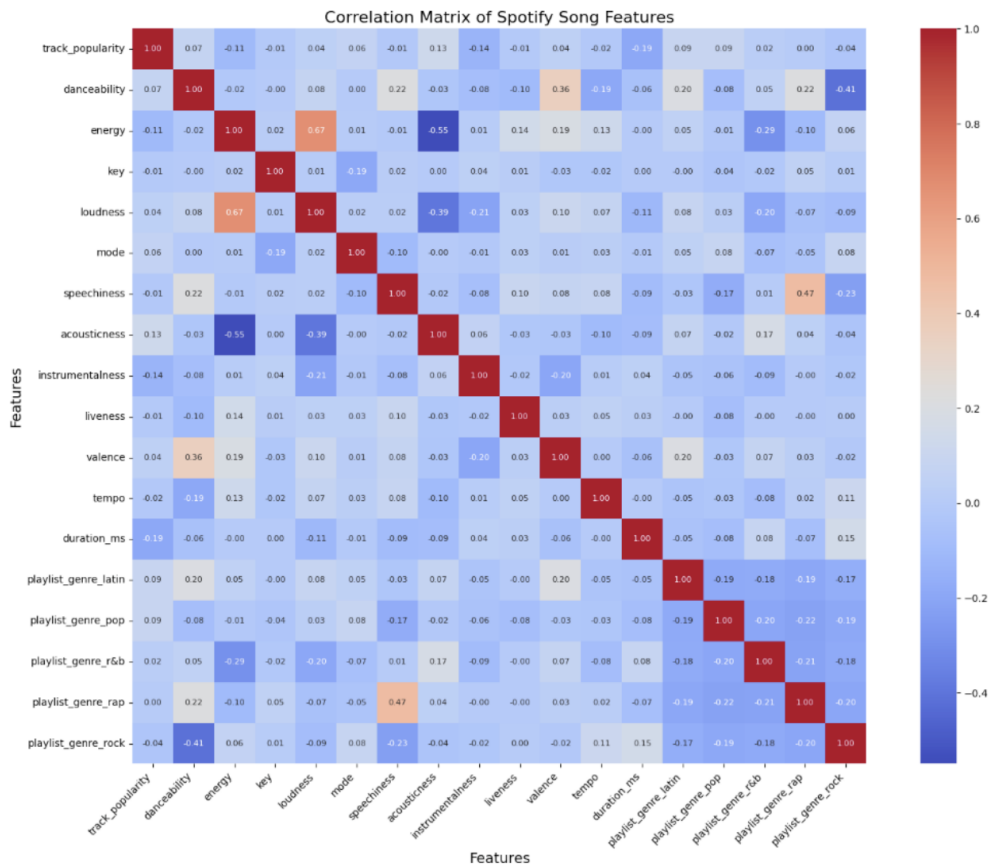

Figure 2: Correlation matrix of Spotify song features.

The first correlation chart highlights the overall correlation values between features in our dataset. As you can see, the darker red values have a stronger positive correlation and darker blues have a stronger negative correlation. Firstly, the primary diagonal axis represents the correlation between a feature and itself, a number that is logically 1. It can be ignored. By setting a threshold magnitude of 0.5, we were able to identify the main features that had a strong correlation:

1. Acousticness and Energy
2. Loudness and Energy

Both pairs logically follow our understanding of the dataset. However, we argue that the omission of any of these three variables from the dataset removes significant information. Energy is a feature that encompasses a number of different factors. While sound and acoustic are critical, it contains many other factors that would be ignored if we were to omit it. Thus, we decided to keep it in the dataset.

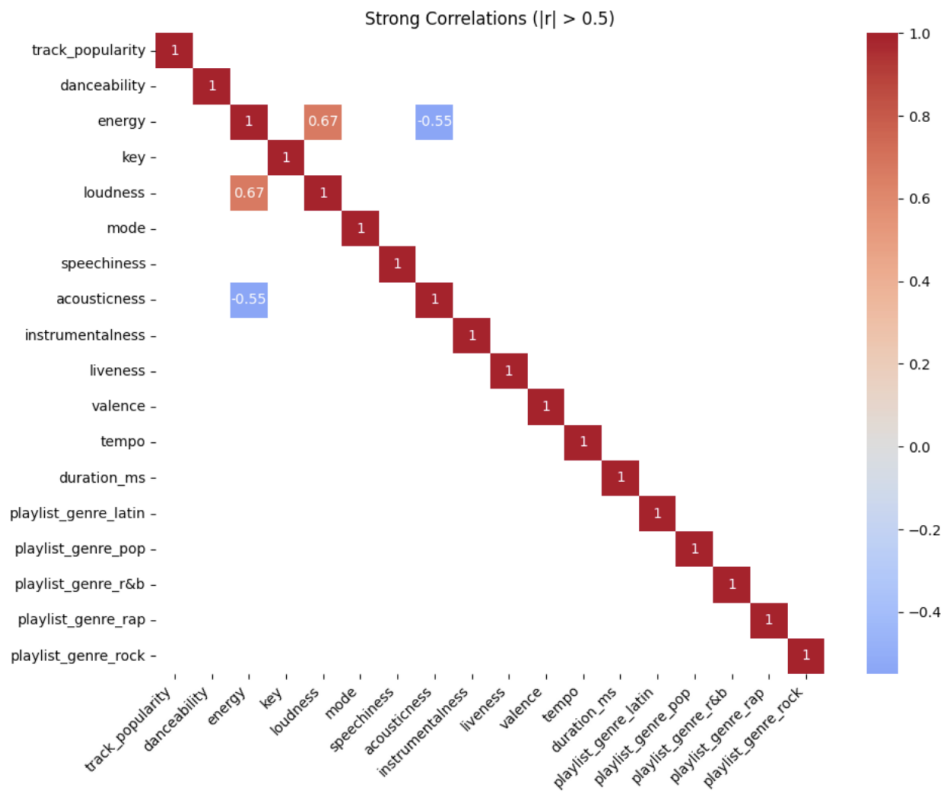

Figure 3: Strong correlations with threshold magnitude greater than 0.5.

5 Question 2: Which predictors are significant in predicting popularity? How suitable is the model for predicting popularity?

We wanted to determine which features of our dataset are useful for predicting the popularity of a song, and also evaluate how suitable our model is to the data we collected. We used several model evaluation techniques like R^2 and adjusted R^2 measures, residual analysis, plots of fitted vs observed values, and K-fold cross-validation. Together, these techniques allowed us to assess the explanatory ability of our model, and its predictive performance on test data.

We trained 3 different models, one with audio features only, one with Genre + audio, and one with all features. We used K-fold cross-validation to choose the best model to ensure that we had a strong sample and that it did not overfit.

M1: Audio Features Only

Features: `[danceability, energy, loudness, acousticness, valence, tempo]`

```
Fold 1: MSE = 688.9632
Fold 2: MSE = 615.2973
Fold 3: MSE = 615.0622
Fold 4: MSE = 563.9561
Fold 5: MSE = 641.0399
CV(5) = 624.8637 (RMSE = 24.9973)
```

M2: Genre + Audio

Features: [danceability, energy, loudness, acousticness, valence, tempo, playlist_genre_latin, playlist_genre_pop, playlist_genre_r&b, playlist_genre_rap, playlist_genre_rock]

```
Fold 1: MSE = 674.6258
Fold 2: MSE = 609.3140
Fold 3: MSE = 592.1733
Fold 4: MSE = 572.4817
Fold 5: MSE = 648.7722
CV(5) = 619.4734 (RMSE = 24.8892)
```

M3: Full Model

Features: [danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalnes
liveness, valence, tempo, duration_ms, playlist_genre_latin, playlist_genre_pop, playlist_genre
playlist_genre_rap, playlist_genre_rock]

```
Fold 1: MSE = 656.6322
Fold 2: MSE = 606.4277
Fold 3: MSE = 604.2645
Fold 4: MSE = 564.8861
Fold 5: MSE = 612.9436
CV(5) = 609.0308 (RMSE = 24.6785)
```

The model which was trained on all features had the lowest RMSE over the 5 folds. This suggests that additional attributes beyond just audio features, and genre are significant in helping predict popularity.

The model with all features M3 is the model that we selected, as it had the lowest CV(5) = 609.0308.

After choosing the model we calculated the AIC, R^2 and Adjusted R^2.

Metric	Value
AIC	9233.2471
R^2	0.1015
Adjusted R^2	0.0859

R^2 is a measure that is the proportion of the variance that is explained by the model, which in this case is 10.15%. After adjusting for the number of predictors, the R^2 value is 8.59%.

Variance Inflation Factors (VIF) measure how much of the variance of a regression coefficient is increased by collinearity. We computed the VIF values to determine which features are highly correlated with each other.

Feature	VIF
energy	2.816728
loudness	2.177307
playlist_genre_rap	2.149657
playlist_genre_r&b	2.081907
playlist_genre_rock	2.049110
playlist_genre_pop	1.849036
playlist_genre_latin	1.769692
danceability	1.626034
acousticness	1.522720
valence	1.463140
speechiness	1.409549
instrumentalness	1.227879
tempo	1.086990
liveness	1.079518
mode	1.078880
duration_ms	1.078563
key	1.045880

(VIF > 10 = multicollinearity concern)

None of these VIF values are above 10, indicating that there is minimal multicollinearity in our model.

We then conducted graph analysis on our residuals and fitted values to determine our model fit with the OLS assumptions.

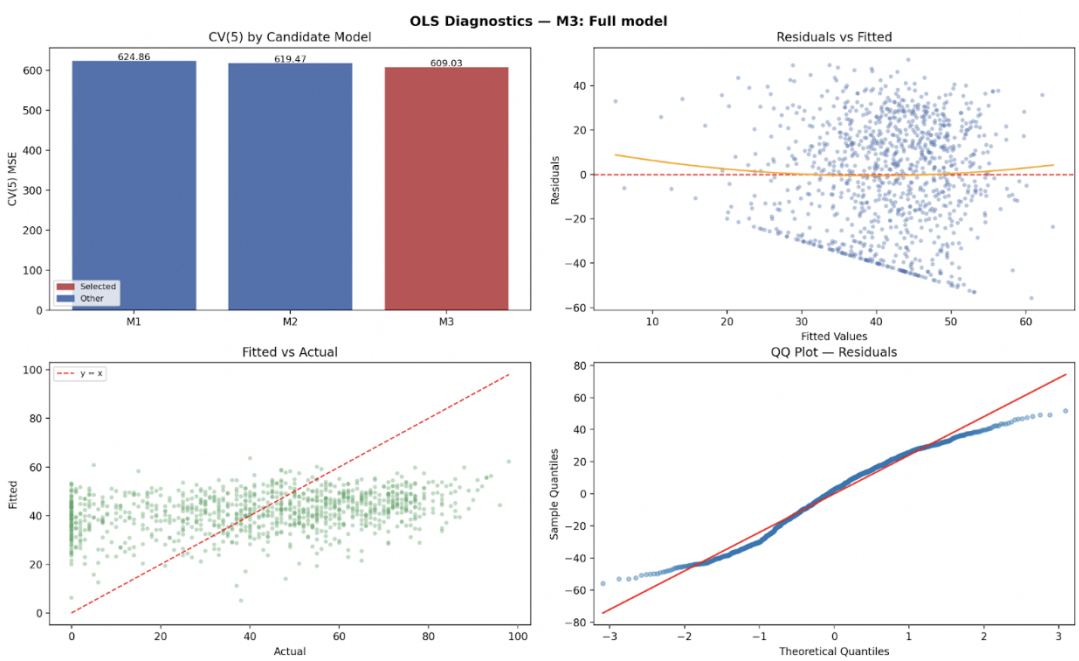

Figure 4: OLS diagnostic plots for the full model.

Residuals vs Fitted

A residuals graph like the one on the top right is used to determine whether the assumptions of linear regression are satisfied. Ideally, residuals should be centered around 0, which we can see is the case, but there also seems to be a negative slope pattern on the lower residuals. This suggests that the model might not have a linear pattern for the outliers.

Fitted vs Actual

The graph on the bottom left shows the model's predictions versus the actual values. As one can see, the fitted values tend to stay within the medium popularity zone, throwing off the correlation, although there is perhaps a slight positive correlation overall.

QQ Plot — Residuals

The QQ Plot — Residuals graph compares the regression residuals distribution with the theoretical quantiles of a normal distribution. This plot is used to evaluate the normality assumption of the residuals in linear regression. If the distribution of the residuals was completely normal, then the points would follow the form of a line. However, the tails of the normal points do not follow a line, indicating that the residuals have heavier tails under a normal distribution, and that there are outliers in the data.

Final Conclusions of This Section

The full regression model including all audio and genre features gave the best predictive performance according to the cross-validation and AIC values obtained. However, the model explains only about 10% of the variability in song popularity, which suggests that popularity is also influenced by external factors beyond the features provided to us in the dataset. On the positive side, multicollinearity is not a big concern, meaning most of our features are not heavily correlated with each other. Ultimately, we have identified the features that can predict popularity well relative to the features we have, but also determined that some of the features that can actually help explain and predict song popularity are not included in our dataset.

6 Question 3: Does the relationship between audio features, such as danceability, valence, energy, etc. and popularity differ across playlist genres?

To investigate whether the relationship between audio features and track popularity varies across playlist genres, we employed Ordinary Least Squares (OLS) multiple linear regression. We used `track_popularity` as the response variable and six audio features as predictors: danceability, energy, loudness, valence, instrumentalness, and duration_ms. Our approach consisted of fitting a single pooled model across all genres, fitting separate per-genre models, and then formally comparing the two using an F-test.

6.1 Pooled Model

We first fit a pooled OLS regression model using all 32,833 songs regardless of genre. This model assumes the relationship between each audio feature and popularity is identical across all genres,

serving as our baseline. The pooled model yielded an adjusted R^2 of 0.0693, and all six predictors were statistically significant ($p < 0.05$).

6.2 Per-Genre Models

We then fit six separate OLS regression models, one for each playlist genre (EDM, Latin, Pop, R&B, Rap, and Rock), each with approximately 5,000 songs. These models allow each genre to have its own coefficients, meaning each genre can exhibit a different relationship between audio features and popularity.

Table 1: OLS Regression Coefficients by Genre

Genre	Danceability	Energy	Loudness	Valence	Instrumental.	Duration
EDM	1.345	-30.794	0.575	7.340	-8.340	-0.0001
Latin	1.940	-50.806	2.940	0.113	-9.487	-0.0000
Pop	15.046	-31.373	2.313	-0.365	-10.354	-0.0000
R&B	5.107	-26.003	2.084	-11.352	-7.213	-0.0001
Rap	22.352	-18.717	0.746	-2.852	-1.795	-0.0000
Rock	11.432	-13.597	0.603	-1.722	-12.352	-0.0000

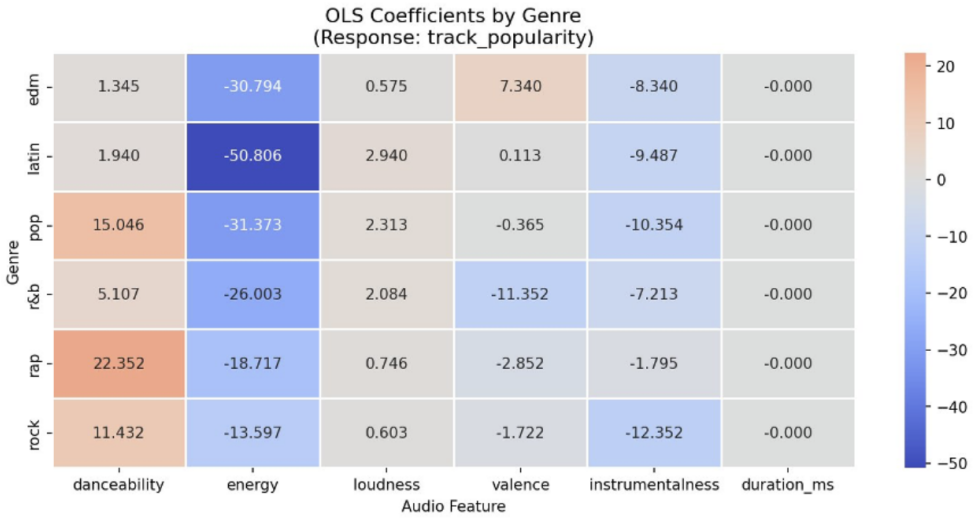

Figure 5: Heatmap of OLS coefficients by genre. Red/warm colors indicate positive coefficients; blue/cool colors indicate negative coefficients.

The coefficients reveal substantial variation across genres. Energy is universally negative, but ranges from -50.8 in Latin to -13.6 in Rock. Danceability has a strong positive effect in Rap ($+22.4$, $p < 0.001$) and Pop ($+15.0$, $p < 0.001$) but is not significant in EDM or Latin. Most notably, valence exhibits opposite effects: positive in EDM ($+7.34$, $p < 0.001$) but negative in R&B (-11.35, $p < 0.001$), suggesting fundamentally different listener preferences regarding emotional tone across genres.

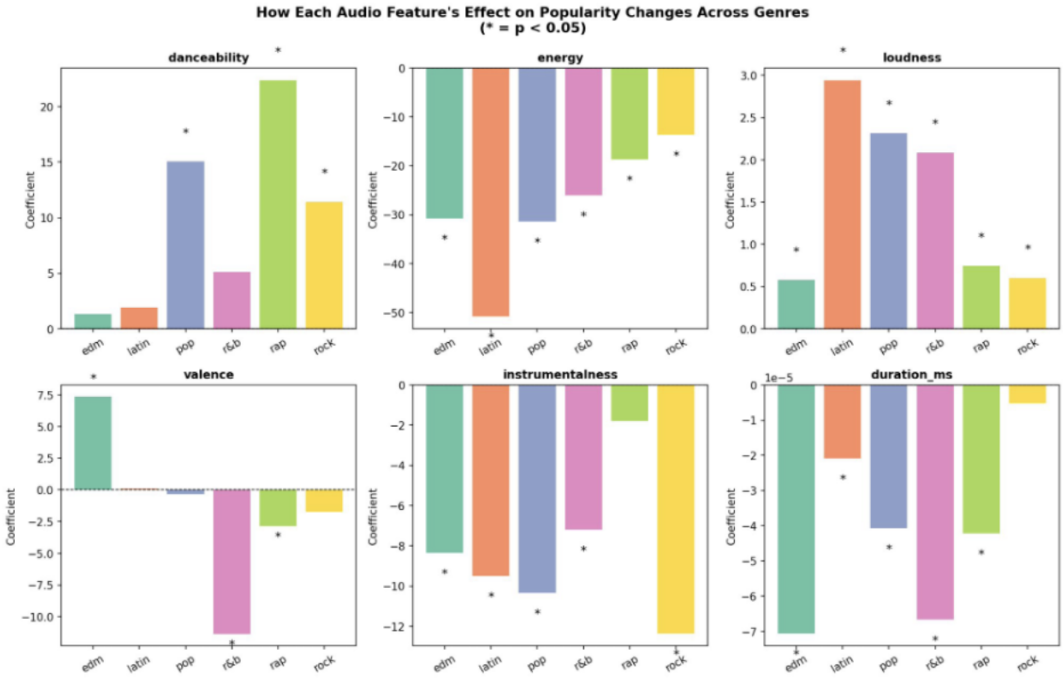

Figure 2: Per-feature bar charts showing how each audio feature's coefficient varies across genres. Asterisks (*) indicate statistical significance at p < 0.05.

Figure 6: Per-feature bar charts showing how each audio feature's coefficient varies across genres. Asterisks indicate statistical significance at $p < 0.05$.

Instrumentalness is negative across all genres, indicating that tracks with more vocal content tend to be more popular regardless of genre. However, the penalty for instrumentalness varies: in Rock, the coefficient is -12.35, while in Rap it is only -1.79 and not statistically significant. Duration has a negligible effect across all genres, with coefficients near zero, suggesting that song length has minimal practical impact on popularity.

6.3 Formal F-Test: Pooled vs. Per-Genre Models

While the coefficient tables suggest differences across genres, we need a formal test to confirm these differences are real and not due to chance. We used an F-test comparing the pooled model (same coefficients for all genres) against the per-genre models (each genre gets its own coefficients). The F-test yielded $F = 36.03$ with $p < 0.0001$, far below our threshold of $\alpha = 0.05$. We reject the null hypothesis and conclude that the per-genre models are significantly better, confirming that the relationship between audio features and popularity does differ across genres.

Table 2: F-Test Results — Pooled vs. Per-Genre Models

Statistic	Value
Pooled Model RSS	19,070,488.89
Per-Genre Models RSS	18,364,239.10
Extra Parameters	35
F-Statistic	36.03
p-value	< 0.0001

6.4 R^2 Analysis

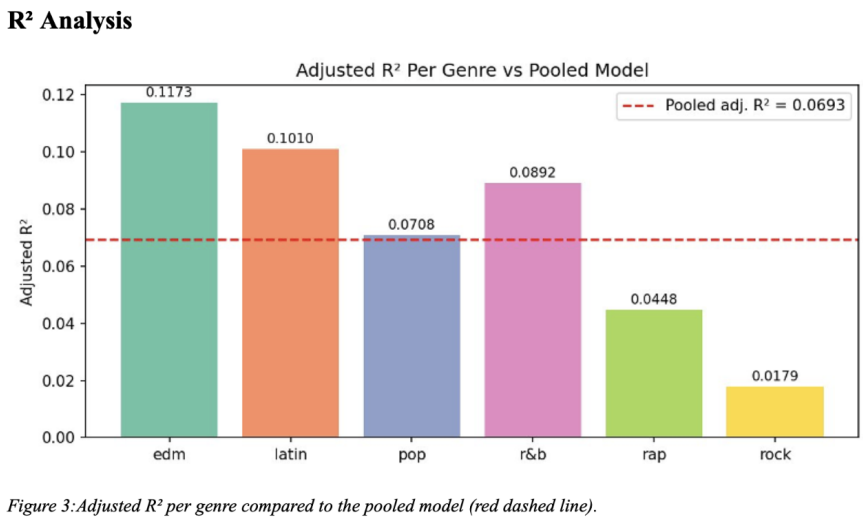

Figure 7: Adjusted R^2 per genre compared to the pooled model (red dashed line).

Figure 3 shows the adjusted R^2 for each genre model. EDM has the highest (0.1173) and Rock the lowest (0.0179). EDM, Latin, and R&B all outperform the pooled model (0.0693), while Rap and Rock fall below it. This suggests audio features matter more for popularity in some genres than others.

6.5 Residual Analysis

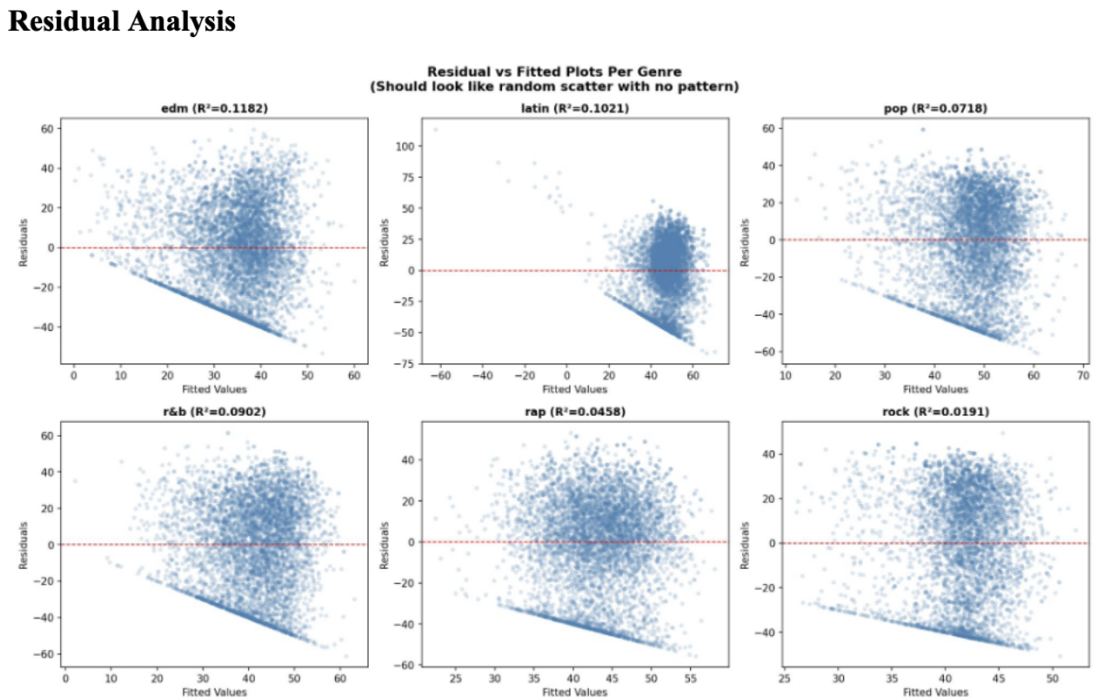

Figure 5: Residual vs. Fitted plots for each per-genre model.

Figure 8: Residual vs. fitted plots for each per-genre model.

The per-genre residual plots show roughly random scatter, though a slight triangular pattern is visible, particularly in EDM and Latin. This suggests some heteroscedasticity, likely due to popularity being bounded between 0 and 100.

6.6 Multicollinearity

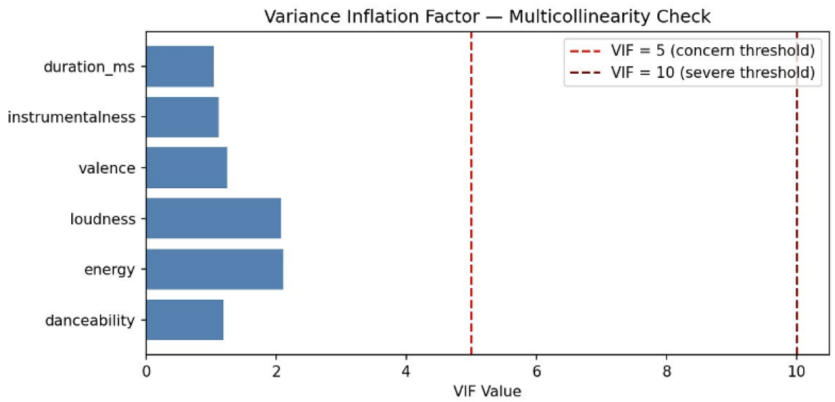

Figure 7: Variance Inflation Factor (VIF) for each predictor.

Figure 9: Variance Inflation Factor (VIF) for each predictor.

Figure 7 presents the VIF values for each audio feature. All VIF values are well below the conventional threshold of 5, with the highest being energy and loudness at approximately 2.2. This is consistent with their pairwise correlation of 0.68, which while moderate, does not pose a severe multicollinearity concern. We can therefore conclude that multicollinearity is not an issue in our model, and the individual coefficient estimates can be interpreted with reasonable confidence.

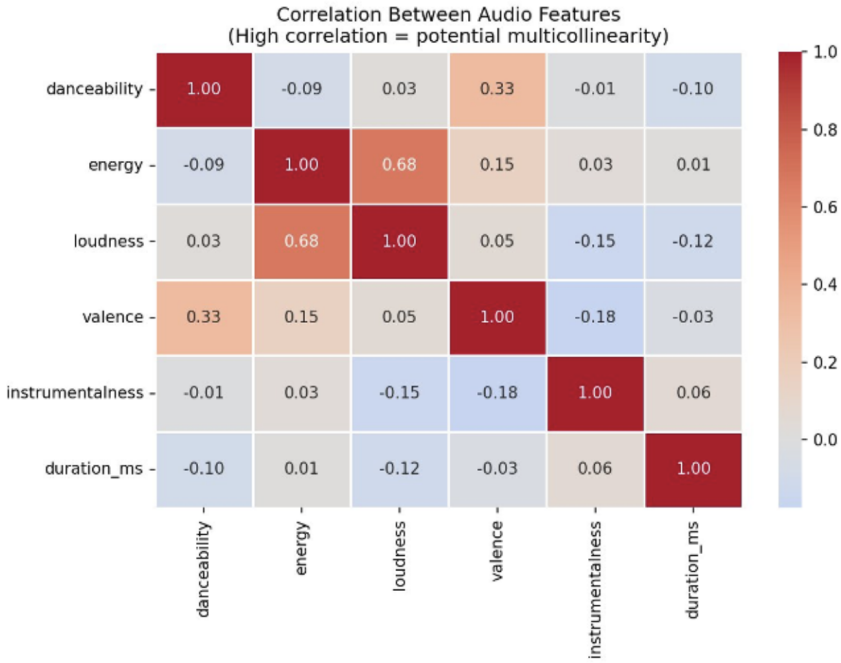

Figure 8: Correlation matrix of audio features. Energy and loudness show the strongest pairwise correlation (0.68).

Figure 10: Correlation matrix of audio features. Energy and loudness show the strongest pairwise correlation (0.68).

6.7 Coefficient Interpretation

Danceability has a strong positive effect in Rap (+22.35), Pop (+15.05), and Rock (+11.43), all with $p < 0.001$, but is not significant in EDM or Latin, likely because nearly all tracks in those genres are already highly danceable. Energy is negative across all genres, meaning more subdued tracks tend to be more popular, but the effect is nearly four times stronger in Latin (−50.81) than in Rock (−13.60). Valence shows the most striking difference: positive in EDM (+7.34) but negative in R&B (−11.35), meaning happy-sounding music is rewarded in EDM but penalized in R&B. Instrumentalness is negative across genres, indicating vocal tracks are more popular, with the largest penalties in Rock (−12.35) and Pop (−10.35).

6.8 Limitations

The low R^2 values (2–12%) indicate audio features alone are weak predictors of popularity, with most variance driven by external factors not in the dataset. The residual plots show heteroscedasticity, which may affect standard error reliability. The energy-loudness correlation (0.68) is moderate but VIF values remain below the threshold of concern.

6.9 Findings

The F-test ($F = 36.03$, $p < 0.0001$) confirms that the relationship between audio features and popularity differs significantly across genres. Each genre has a distinct "formula": Rap rewards danceability, EDM rewards happy-sounding music, R&B penalizes it, and energy is universally negative but varies in strength. While the models have limited explanatory power, the significant coefficients reveal meaningful structural differences across genres, suggesting any predictive model would benefit from genre-specific approaches rather than a one-size-fits-all model.

7 Conclusion and Implications

Ultimately, our analysis shows that audio features such as energy, loudness, valence and danceability have statistically significant relationships with song popularity. However, in our findings we learned that their overall explanatory power is relatively limited. The best performing model trained only explained a small proportion of variability in song popularity, which suggests that there are other features not included in this dataset that are needed to predict popularity. However, our analysis does reveal several meaningful differences across genres. The effect of audio features changes based on the playlist genre, and the formal F-test we conducted confirms that models that are genre specific are a better first for the data and perform better than models that are single pooled. Thus, this implies that popularity is not determined by a universal set of musical characteristics, but also depends on genre-specific factors.